# Kurt Matthew Nudo

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## WORK EXPERIENCE

**International Business Machines (IBM)** **Feb. 2026 – Present**
*Experienced Full Stack Developer for Banking and Financial Markets, Japanese Bilingual* *Quezon City, NCR*

- Deliver backend modules for a large-scale Japanese banking migration, using Java 17, Jakarta EE, Maven, MyBatis, Oracle, JUnit, and Mockito.
- Draft initial module implementations from reviewed Japanese design documents, then integrate, inspect, debug, and refine them by hand before release.
- Translate Japanese technical specifications into production-ready business logic, contributing to 20+ migrated modules while maintaining financial and transaction integrity.
- Write and maintain 50+ unit tests by hand, covering test data, database setup, exception handling, and verification of business logic and database interactions.
- Diagnose and resolve 20+ defects and edge cases through code-level analysis and database verification, reducing integration and release issues.
- Apply DevOps practices across automated testing, CI builds, deployment, and release validation.

**Advanced World Solutions** **Jul. 2024 – Feb. 2026**
*Full Stack Developer, Systems Migration Engineer (Legacy C# System → React + TypeScript, Node.js)* *Makati, NCR*

- Participated in a long-term migration project transforming a legacy C# desktop system for a Japanese client into a modern web architecture using TypeScript, React, and Node.js.
- Achieved >95% feature parity and codebase modernization, introducing modular architecture and asynchronous data handling.
- Collaborated directly with Japanese engineers to clarify technical specifications, resolve ambiguous requirements, and ensure language-accurate system translation.
- Migrated over 20 production modules from a legacy C# desktop system to a TypeScript, React, and Node.js web stack for a Japanese enterprise client within the first year.
- Integrated Oracle PL/SQL through REST APIs for data persistence, query optimization, and transaction management within the migrated web modules.
- Closed over 40 verified bug-fix tickets, resolving defects across UI workflows, API integrations, and localized Japanese content.
- Developed and maintained unit and integration tests using Jest for each migrated module, ensuring ≥90% code coverage as a release criterion.

## PROJECTS

**Typhoon Replay** **2026**
*Web app: replays of typhoons that hit the Philippines or Japan since 1940* *typhoon.krtmtthw.dev*

- Hour-by-hour ERA5 reanalysis beside the official best track, in English and Japanese. **Technologies**: React, TypeScript, Java 17, Spring Boot, PostgreSQL/PostGIS, Python, GitHub Actions

**Lights** **2026**
*Web app: poverty estimates for every Philippine city and municipality, 2012–2025* *lights.krtmtthw.dev*

- PSA official figures plus labelled model estimates with calibrated uncertainty. **Technologies**: Rust, WebAssembly, Python, LightGBM, Google Earth Engine, React, MapLibre

## PUBLISHED LITERATURE

**Temporal Analysis of Nighttime Lights and Vegetation Indices for Poverty Forecasting: A Recurrent Neural Network Perspective**

- Co-author of a study on remote sensing methodologies for poverty forecasting using vegetation indices and nighttime light intensity as key variables (Springer, 2026). Extracted data from Google Earth Engine and government databases, architected model design, and led analysis using recurrent neural networks.
- **Technologies**: Python, TensorFlow, JavaScript for Earth Engine

## EDUCATION

**Saint Louis University** **Jul. 2024**
*Bachelor of Science in Computer Science* *Baguio, Benguet*

- Graduated Summa Cum Laude with a general average of 93.

**Japanese-Language Proficiency Test (JLPT) N4** · *studying for N3* **Certified**